Supplementary material for the breast ultrasound segmentation study

Transfer of fixed review thresholds for breast ultrasound segmentation quality control

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Table S1. Locked thresholds and scoring settings

Setting	Value
Development budget	20 % target. Each method flagged 43/213 (20.19 %) in development.
Locked risk threshold A	0.13705134391784668
Locked risk threshold B	0.06972856653991777
Review trigger	Risk score ≥ the corresponding locked threshold.
Fixed threshold ties	Include every case at the cutoff. If two cutoffs are equally close to the target, select the one with fewer reviews.
SAM source commit	dca509fe793f601edb92606367a655c15ac00fdf
Checkpoint SHA-256	ec2df62732614e57411cdcf32a23ffdf28910380d03139ee0f4fcbe91eb8c912
Formal scoring environment	Python 3.8.20, PyTorch 2.4.1+cu118, torchvision 0.19.1+cu118, Pillow 10.4.0, NumPy 1.24.4.

SAM, Segment Anything Model. Inference used the official sam_vit_b_01ec64.pth checkpoint with float32 weights on a GeForce RTX 3060 Laptop GPU. Statistical replay used Python 3.12.14.

Sensitivity to the low quality definition

Table S2. Test performance at alternative reference Dice cutoffs

Dice cutoff	Test cohort	Method	Detection, n/N (%)	Simulated review, n/N (%)	Misses, n	Misses per 100 patients
< 0.50	BUS-BRA internal	A	52/53 (98.11 %)	174/851 (20.45 %)	1	0.12
< 0.50	BUS-BRA internal	B	43/53 (81.13 %)	150/851 (17.63 %)	10	1.18
< 0.50	BrEaST external	A	5/7 (71.43 %)	35/252 (13.89 %)	2	0.79
< 0.50	BrEaST external	B	5/7 (71.43 %)	33/252 (13.10 %)	2	0.79
< 0.80	BUS-BRA internal	A	109/183 (59.56 %)	174/851 (20.45 %)	74	8.70
< 0.80	BUS-BRA internal	B	95/183 (51.91 %)	150/851 (17.63 %)	88	10.34
< 0.80	BrEaST external	A	21/50 (42.00 %)	35/252 (13.89 %)	29	11.51
< 0.80	BrEaST external	B	21/50 (42.00 %)	33/252 (13.10 %)	29	11.51

Both risk thresholds remained fixed. Detection uses the low-quality masks under the stated cutoff as its denominator. Simulated review and misses per 100 use all scored patients.

Detection rate contrasts

Table S3. Detection rate contrasts at alternative reference Dice cutoffs

Dice cutoff	Contrast	Difference, pp	Nominal 95 % CI, pp
< 0.50	Method A detection, external minus internal	−26.68	[−62.25, −4.80]
< 0.50	External detection, Method B minus Method A	0.00	[−27.14, +27.14]
< 0.80	Method A detection, external minus internal	−17.56	[−31.92, −2.01]
< 0.80	External detection, Method B minus Method A	0.00	[−10.06, +10.06]

pp, percentage points. CI, confidence interval. Between-cohort contrasts use independent Newcombe intervals. Within-BrEaST contrasts use paired Newcombe intervals.

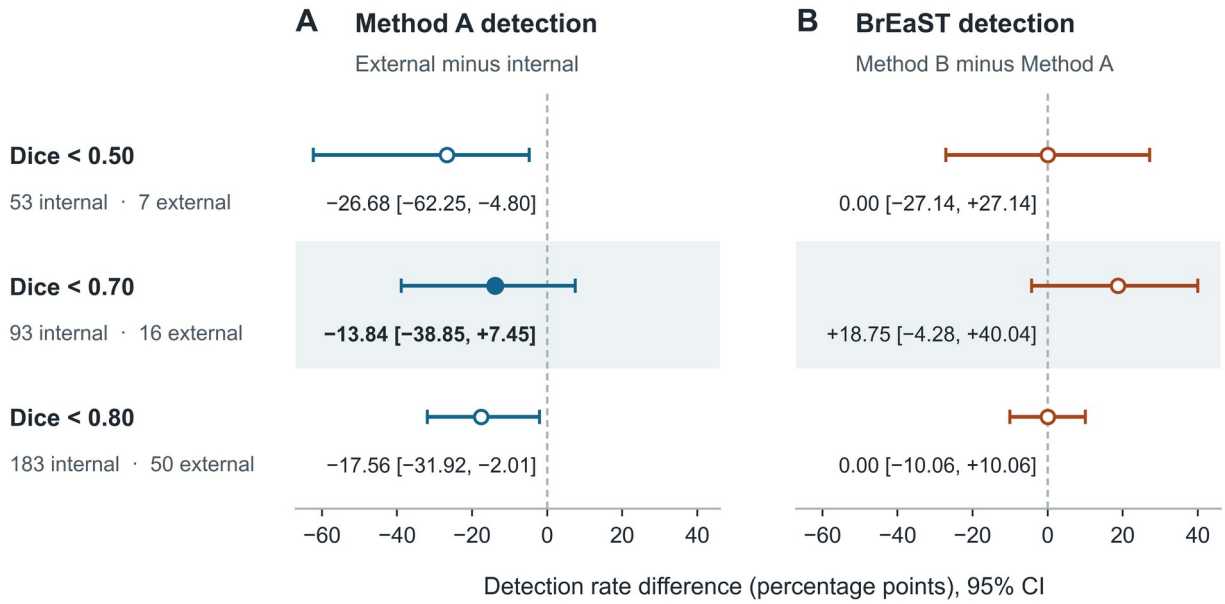


Figure S1. Sensitivity to the definition of low quality. Panel A shows the external-minus-internal detection difference for method A. Panel B shows the paired B-minus-A difference in BrEaST. Risk thresholds remain fixed while reference Dice cutoffs vary. The filled marker denotes H1 at Dice below 0.70. The remaining 95 % confidence intervals are nominal.

Equal review volume in BrEaST

Table S4. BrEaST performance at equal review volume

Method	Detection, n/N (%)	Simulated review, n/N (%)	Misses, n	Misses per 100 patients
A	14/16 (87.50 %)	50/252 (19.84 %)	2	0.79
B	13/16 (81.25 %)	50/252 (19.84 %)	3	1.19

Each method selected 50 of the 252 scored cases in descending risk order. Ties followed lexicographic case identifier order. Low quality used reference Dice below 0.70.

Among the 16 low-quality cases, both methods flagged 12, only A flagged 2, only B flagged 1, and neither flagged 1. The paired B-minus-A difference was −6.25 percentage points (nominal 95 % confidence interval [−31.61, +19.57]). This interval is conditional on the observed rankings.

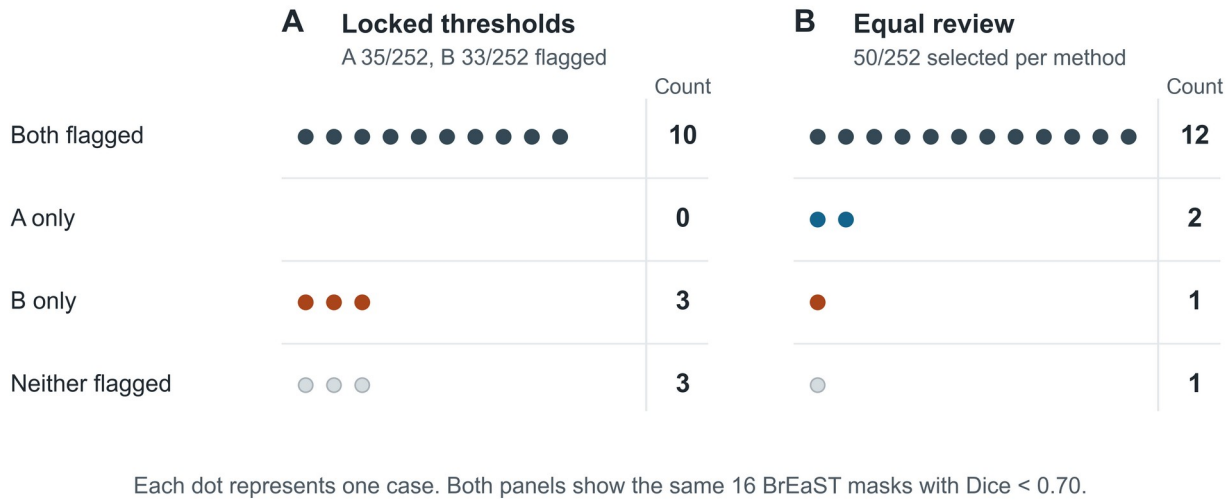


Figure S2. Paired outcomes among the 16 low-quality BrEaST masks. Each dot represents one low-quality case. Panel A uses the two locked thresholds. Panel B ranks each score separately and selects 50 of 252 patients per method. The right column in each panel reports the counts.

Computational reproducibility

Table S5. Agreement between the raw-image rerun and archived scores

Cohort	Compared / expected	Maximum absolute difference	Cases with changed flags, n	Cases with changed status, n
BUS-BRA development	213/213	0	0	0
BUS-BRA internal	851/851	0	0	0
BrEaST external	252/252	0	0	0

The numerical comparison covered reference Dice, SAM predicted intersection over union, the two risk scores and the four shifted-mask Dice values. Flag and count comparisons included the quality label, both review flags, the empty-mask flag and the distinct-box count. Runtime was recorded separately.

Recomputing statistics from archived scores and rerunning inference from the original images both reproduced the locked thresholds, primary and supplementary results, and review-curve counts. The reproducibility package contains the case manifests, input hashes and per-cohort verification records.

Revision audit provenance. The present revision re-executed statistics from archived scores and separately from the previously saved raw-image rerun scores. It did not rerun SAM inference. The study's original same-environment image rerun is documented by its archived receipt and comparisons in Table S5. Computational reproducibility is distinct from independent replication.

Additional descriptive quality-control measures

These post hoc descriptive analyses use the unchanged locked risk thresholds and the primary reference Dice cutoff of 0.70. The positive condition is a low-quality segmentation; the positive test is a review flag. These are mask-quality screening measures, not breast-cancer diagnostic measures. Development results describe calibration data and are not independent test performance.

Table S6. Classification counts at the primary quality cutoff

Cohort	Score	TP, n	FP, n	TN, n	FN, n
BUS-BRA development	A	19	24	165	5
BUS-BRA development	B	17	26	163	7
BUS-BRA internal	A	71	103	655	22
BUS-BRA internal	B	68	82	676	25
BrEaST external	A	10	25	211	6
BrEaST external	B	13	20	216	3

TP, true positive (flagged low quality); FP, false positive (flagged quality-acceptable); TN, true negative (unflagged quality-acceptable); FN, false negative (unflagged low quality); n, number of patients. A, model-confidence risk; B, prompt-disagreement risk.

Table S7. Sensitivity, specificity and predictive values

Cohort	Score	Sensitivity	Specificity	PPV	NPV
BUS-BRA development	A	19/24 (79.17 %) [59.53, 90.76]	165/189 (87.30 %) [81.80, 91.32]	19/43 (44.19 %) [30.43, 58.89]	165/170 (97.06 %) [93.30, 98.74]
BUS-BRA development	B	17/24 (70.83 %) [50.83, 85.09]	163/189 (86.24 %) [80.61, 90.44]	17/43 (39.53 %) [26.37, 54.42]	163/170 (95.88 %) [91.75, 97.99]
BUS-BRA internal	A	71/93 (76.34 %) [66.77, 83.83]	655/758 (86.41 %) [83.79, 88.67]	71/174 (40.80 %) [33.78, 48.23]	655/677 (96.75 %) [95.13, 97.84]
BUS-BRA internal	B	68/93 (73.12 %) [63.32, 81.08]	676/758 (89.18 %) [86.77, 91.20]	68/150 (45.33 %) [37.58, 53.32]	676/701 (96.43 %) [94.79, 97.57]
BrEaST external	A	10/16 (62.50 %) [38.64, 81.52]	211/236 (89.41 %) [84.83, 92.72]	10/35 (28.57 %) [16.33, 45.05]	211/217 (97.24 %) [94.10, 98.73]
BrEaST external	B	13/16 (81.25 %) [56.99, 93.41]	216/236 (91.53 %) [87.27, 94.45]	13/33 (39.39 %) [24.68, 56.32]	216/219 (98.63 %) [96.05, 99.53]

Entries show n/N (%), followed by Wilson 95 % confidence intervals in percent. PPV, positive predictive value; NPV, negative predictive value. Sensitivity equals the low-quality detection measure used in the prespecified analysis. Predictive values depend on the frequency of low-quality segmentations. Machine-readable descriptive results for Dice cutoffs 0.50, 0.70 and 0.80 are supplied as Additional_quality_control_metrics.csv. Misses per 100 patients remain in the original main Table 2 and supplementary Tables S2 and S4.

Exploratory diagnostic stratification

Table S8. Quality-control detection and review by dataset-reported benign/malignant classification

Cohort	Diagnostic stratum	N / low quality	Score	Detection, n/F (%) [95 % CI, %]	Review, n/N (%) [95 % CI, %]
BUS-BRA internal	Benign	577 / 53	A	51/53 (96.23 %) [87.25, 98.96]	102/577 (17.68 %) [14.78, 21.00]
BUS-BRA internal	Benign	577 / 53	B	41/53 (77.36 %) [64.47, 86.55]	80/577 (13.86 %) [11.28, 16.92]
BUS-BRA internal	Malignant	274 / 40	A	20/40 (50.00 %) [35.20, 64.80]	72/274 (26.28 %) [21.42, 31.79]
BUS-BRA internal	Malignant	274 / 40	B	27/40 (67.50 %) [52.02, 79.92]	70/274 (25.55 %) [20.75, 31.02]
BrEaST external	Benign	154 / 8	A	5/8 (62.50 %) [30.57, 86.32]	15/154 (9.74 %) [5.99, 15.45]
BrEaST external	Benign	154 / 8	B	7/8 (87.50 %) [52.91, 97.76]	13/154 (8.44 %) [5.00, 13.91]
BrEaST external	Malignant	98 / 8	A	5/8 (62.50 %) [30.57, 86.32]	20/98 (20.41 %) [13.62, 29.43]
BrEaST external	Malignant	98 / 8	B	6/8 (75.00 %) [40.93, 92.85]	20/98 (20.41 %) [13.62, 29.43]

Post hoc exploratory analysis with unchanged locked thresholds and reference Dice < 0.70. N, all scored patients in the stratum; F, low-quality masks; n, detected or reviewed cases as indicated; CI, Wilson confidence interval; A, model-confidence score; B, prompt-disagreement score. Classification is dataset-reported and is not uniformly histopathologically confirmed in BrEaST. These are quality-screening measures, not diagnostic sensitivity for malignancy. No hypothesis or interaction tests were performed. Complete machine-readable results also include the development strata, confusion counts and misses per 100 patients.

Empty-mask audit

Across 213 development, 851 internal and 252 external images, the saved initial_mask_empty flag was zero for every initial prediction. Every one of the 5,264 initial-to-shifted Dice values was strictly positive, which also establishes that no shifted prediction was empty. Consequently the convention Dice(empty, empty)=1 affected zero recorded scores. A reference Dice of zero alone was not treated as evidence of an empty prediction. No affected-case sensitivity analysis was needed.

External missed-case records

Table S9. All six low-quality external cases missed by method A

Case / label	Detected by	Reference Dice	Risk A	A flag	Risk B	B flag	Mean consistency Dice
case089 malignant	Neither	0.3191	0.104072	0	0.041041	0	0.9590
case175 benign	Neither	0.0293	0.114944	0	0.020059	0	0.9799
case238 malignant	Neither	0.6813	0.038909	0	0.005766	0	0.9942
case024 benign	B only	0.6852	0.054510	0	0.109995	1	0.8900
case105 malignant	B only	0.6546	0.102471	0	0.259671	1	0.7403
case222 benign	B only	0.5784	0.126987	0	0.201743	1	0.7983

All external cases with reference Dice < 0.70 and A flag=0 are included, partitioned by the B flag. A flag and B flag equal 1 when risk is at least the respective full-precision locked cutoff: A=0.13705134391784668; B=0.06972856653991777. Displayed scores are rounded; decisions use the original precision. Mean consistency is one minus Risk B. Each initial prediction was nonempty, and every initial-to-shifted Dice was positive. This outcome-defined case set illustrates score behavior and is not used to estimate an additional performance advantage.

The original six initial SAM masks were not retained in the accessible archive. The accompanying case-input visual aid contains only released images and manual reference masks, clearly labelled as such; no predicted contour was synthesized. A targeted replay script is supplied to recover model predictions once the exact checksum-verified checkpoint is available. The existing Figure 2 case001 is detected by both scores and is not substituted for any of these six cases.